

AI-POWERED INDIAN CYBERCRIME & THREAT INTELLIGENCE ANALYTICS PLATFORM

ABSTRACT

The rapid growth of digital platforms, online banking systems, e-commerce applications, and internet services has increased cybercrime activities across India. Cyber attacks such as phishing, ransomware, malware, identity theft, online fraud, hacking, and data breaches are causing significant financial losses to individuals, organizations, and public sectors.

This project focuses on analyzing Indian cybercrime incidents using Python, SQL Server, Power BI, and AI integration. The system performs complete data cleaning and preprocessing, SQL-based threat intelligence analysis, interactive dashboard development, and AI-powered cybersecurity insights using a Streamlit assistant.

The final output is a professional cybercrime intelligence platform that helps users identify high-risk attack types, city-level cybercrime patterns, sector-wise financial losses, yearly trends, and important threat intelligence insights.

ABOUT THE DATASET

The dataset contains structured cybercrime incident records. Each record represents a cybercrime case and includes details such as year, day, amount lost in INR, incident type, city, and affected category.

Attribute	Description
Dataset Type	Cybersecurity / Cybercrime Dataset
Dataset Nature	Structured Tabular Dataset
Dataset Level	Cybercrime Incident Level
Total Records	1200 Cybercrime Cases
Main Target	Cybercrime trend, risk, and financial loss analysis

Data Mix

- Numerical fields: year, day, amount_lost_inr, high-risk indicator.
- Categorical fields: incident_type, city, category, amount_loss_category, day_range.
- Derived fields: amount_lost_category, is_high_value_attack, day_range, case_date.

DOMAIN BACKGROUND

Cybercrime is one of the most important digital threats in India. With the rise of online banking, digital payments, e-commerce, cloud applications, and social media platforms, attackers use different methods to target users and organizations.

Common cybercrime types include ransomware, phishing, malware, online fraud, identity theft, data breaches, cyber bullying, and hacking. These attacks create financial losses, privacy issues, data theft, and operational risks.

A cybercrime analytics platform helps convert raw incident data into meaningful intelligence. It supports decision-making by showing which attack types, cities, sectors, and time periods are more vulnerable.

WHY THIS DATA IS IMPORTANT

- Helps monitor cybercrime growth patterns.
- Identifies high-risk cyber attack types.
- Shows financial loss contribution by city, sector, and attack category.
- Supports cybersecurity awareness and digital safety decisions.
- Helps analysts build threat intelligence reports from structured data.
- Provides a base for future AI, forecasting, and risk prediction models.

BUSINESS PROBLEM

Main Problem: How can Indian cybercrime data be analyzed effectively to understand cyber attack patterns, financial losses, high-risk threats, and sector-level vulnerabilities?

Problem Breakdown

- Increasing cybercrime cases require trend analysis across years.
- Financial loss monitoring is needed to identify severe cyber incidents.
- City-level risk analysis helps identify highly affected locations.
- Sector-wise analysis shows which domains face more cyber threats.
- AI-powered insights help users ask questions and understand the dataset interactively.

DATA DICTIONARY

Column Name	Description
year	Year in which the cybercrime case occurred
day	Day of the month
amount_lost_inr	Financial loss amount in Indian Rupees
incident_type	Type of cyber attack such as ransomware or phishing
city	Indian city where the case was recorded
category	Affected sector/category
amount_lost_category	Low, medium, or high financial loss group
is_high_value_attack	Binary indicator for high-risk/high-value attack
day_range	Early, mid, or late month segmentation
case_date	Generated date field for analysis

EDA – EXPLORATORY DATA ANALYSIS AND PYTHON CLEANING

Python and Pandas were used to clean the raw cybercrime dataset. The cleaning pipeline focused on importing data, renaming columns, removing duplicates, cleaning text columns, fixing spelling issues, handling datatypes, managing missing values, handling outliers, creating new features, and validating the final dataset.

Step 1 – Import Dataset

Imported Pandas and NumPy, loaded the cybercrime CSV file, checked original shape, displayed first rows, and inspected data types. The dataset originally contained 1200 rows and 6 columns.

```
import pandas as pd
import numpy as np
```

```
# =====
# STEP 1: IMPORT DATASET
# =====
df = pd.read_csv("cybersecurity_cases_india_combined.csv")
print("Original Shape :", df.shape)
df.head()
df.info()
```

```
Original Shape : (1200, 6)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1200 entries, 0 to 1199
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Year                  1200 non-null  int64
1   Day                   1200 non-null  int64
2   Amount_Lost_INR       1200 non-null  int64
3   Incident_Type         1200 non-null  object
4   City                  1200 non-null  object
5   Category              1200 non-null  object
dtypes: int64(3), object(3)
memory usage: 56.4+ KB
```

Step 1 – Import Dataset

Step 2 – Rename Columns for SQL / Power BI

Column names were standardized by stripping spaces, converting to lowercase, and replacing spaces with underscores. This makes the dataset clean, SQL-friendly, and easy to use in Power BI.

```
# =====  
# STEP 2: RENAME COLUMNS FOR SQL / POWER BI  
# =====  
  
df.columns = (  
    df.columns  
    .str.strip()  
    .str.lower()  
    .str.replace(" ", "_")  
)  
  
print("Columns after rename:")  
print(df.columns.tolist())
```

Columns after rename:

```
['year', 'day', 'amount_lost_inr', 'incident_type', 'city', 'category']
```

Step 2 – Rename Columns for SQL / Power BI

Step 3 – Remove Duplicates

Duplicate rows were checked and removed. The output showed zero duplicate rows, which means no duplicate cybercrime records were present.

```
# =====  
# STEP 3: REMOVE DUPLICATES  
# =====  
  
duplicates = df.duplicated().sum()  
print("Duplicate rows:", duplicates)  
  
df = df.drop_duplicates().reset_index(drop=True)
```

Duplicate rows: 0

Step 3 – Remove Duplicates

Step 4 – Clean Text Columns

Text columns were cleaned by removing extra spaces and converting text to title case. This improves consistency in city names, categories, and attack types.

```
# =====  
# STEP 4: CLEAN TEXT COLUMNS  
# =====  
  
text_columns = df.select_dtypes(include=["object"]).columns  
  
for col in text_columns:  
    df[col] = df[col].str.strip().str.title()
```

Step 4 – Clean Text Columns

Step 5 – Fix Spelling / Category Issues

Known spelling and category inconsistencies were corrected, such as Bangalore to Bangalore, Cyber Fraud to Online Fraud, and E Commerce to E-Commerce.

```
# =====  
# STEP 5: FIX KNOWN SPELLING / CATEGORY ISSUES  
# =====  
  
if "incident_type" in df.columns:  
    df["incident_type"] = df["incident_type"].replace({  
        "Malware_Attacks": "Malware",  
        "Malware Attack": "Malware",  
        "Phising": "Phishing",  
        "Cyber Fraud": "Online Fraud",  
        "Financial Fraud": "Online Fraud"  
    })  
  
if "city" in df.columns:  
    df["city"] = df["city"].replace({  
        "Banglore": "Bangalore",  
        "Bengaluru": "Bangalore",  
        "Hyd": "Hyderabad",  
        "Del": "Delhi",  
        "New Delhi": "Delhi"  
    })  
  
if "category" in df.columns:  
    df["category"] = df["category"].replace({  
        "Govt": "Government",  
        "Edu": "Education",  
        "Ecommerce": "E-Commerce",  
        "E Commerce": "E-Commerce"  
    })
```

Step 5 – Fix Spelling / Category Issues

Step 6 & 7 – Fix Data Types and Handle Missing Values

Year and day columns were converted to integer type, amount lost was converted to numeric, and missing values were handled using median for numeric columns and Unknown for categorical columns.

```
# =====  
# STEP 6: FIX DATA TYPES  
# =====  
  
if "year" in df.columns:  
    df["year"] = pd.to_numeric(df["year"], errors="coerce").astype("Int64")  
  
if "day" in df.columns:  
    df["day"] = pd.to_numeric(df["day"], errors="coerce").astype("Int64")  
  
if "amount_lost_inr" in df.columns:  
    df["amount_lost_inr"] = pd.to_numeric(df["amount_lost_inr"], errors="coerce")  
  
# =====  
# STEP 7: HANDLE MISSING VALUES  
# =====  
  
for col in df.columns:  
    if df[col].isnull().sum() > 0:  
        if pd.api.types.is_numeric_dtype(df[col]):  
            df[col] = df[col].fillna(df[col].median())  
        else:  
            df[col] = df[col].fillna("Unknown")
```

Step 6 & 7 – Fix Data Types and Handle Missing Values

Step 8 – Handle Outliers

The IQR method was applied to identify and control extreme financial loss values. This avoids distortion in financial analysis.

```
# =====  
# STEP 8: HANDLE OUTLIERS  
# =====  
  
if "amount_lost_inr" in df.columns:  
    q1 = df["amount_lost_inr"].quantile(0.25)  
    q3 = df["amount_lost_inr"].quantile(0.75)  
    iqr = q3 - q1  
  
    lower_limit = q1 - 1.5 * iqr  
    upper_limit = q3 + 1.5 * iqr  
  
    df["amount_lost_inr"] = df["amount_lost_inr"].clip(  
        lower=lower_limit,  
        upper=upper_limit  
    )
```

Step 8 – Handle Outliers

Step 9 – Feature Engineering

New analytical columns were created: `amount_lost_category`, `is_high_value_attack`, `day_range`, and `case_date`. These fields support dashboard visuals and SQL analysis.

```
# =====  
# STEP 9: FEATURE ENGINEERING  
# =====  
  
if "amount_lost_inr" in df.columns:  
    df["amount_lost_category"] = pd.qcut(  
        df["amount_lost_inr"],  
        q=3,  
        labels=["Low", "Medium", "High"],  
        duplicates="drop"  
    )  
  
    high_value_threshold = df["amount_lost_inr"].quantile(0.75)  
  
    df["is_high_value_attack"] = np.where(  
        df["amount_lost_inr"] >= high_value_threshold,  
        1,  
        0  
    )  
  
if "day" in df.columns:  
    df["day_range"] = pd.cut(  
        df["day"],  
        bins=[0, 10, 20, 31],  
        labels=["Early Month", "Mid Month", "Late Month"],  
        include_lowest=True  
    )  
  
if "year" in df.columns and "day" in df.columns:  
    df["case_date"] = pd.to_datetime(  
        df["year"].astype(str) + "-01-" + df["day"].astype(str),  
        errors="coerce"  
    )
```

Step 9 – Feature Engineering

Step 10 – Final Validation and Export

Final checks confirmed 1200 rows, 10 columns, zero missing values, zero duplicate rows, and correct final datatypes. The cleaned dataset was exported successfully.

```
# -----  
# STEP 10: FINAL VALIDATION  
# -----  
  
print("\nFinal Shape:", df.shape)  
print("\nMissing Values:")  
print(df.isnull().sum())  
  
print("\nDuplicate Rows:", df.duplicated().sum())  
  
print("\nFinal Data Types:")  
print(df.dtypes)  
  
print("\nFinal Preview:")  
print(df.head())  
  
Final Shape: (1200, 10)  
  
Missing Values:  
year          0  
day           0  
amount_lost_inr  0  
incident_type  0  
city          0  
category      0  
amount_lost_category  0  
is_high_value_attack  0  
day_range      0  
case_date      0  
dtype: int64  
  
Duplicate Rows: 0  
  
Final Data Types:  
year          Int64  
day           Int64  
amount_lost_inr  int64  
incident_type  object  
city          object  
category      object  
amount_lost_category  category  
is_high_value_attack  int64  
day_range      category  
case_date      datetime64[ns]  
dtype: object  
  
Final Preview:  
   year  day  amount_lost_inr  incident_type  city  category  
0  2022   22         86630      Data Breach  Bangalore  Corporate  
1  2023   21         231983       Phishing  Kolkata  Educational  
2  2021    6         284707      Data Breach  Bangalore  E-Commerce  
3  2023    6         433557       Phishing  Ahmedabad   Health  
4  2023   23         378816       Phishing   Jaipur  Government  
  
   amount_lost_category  is_high_value_attack  day_range  case_date  
0                Low          0  Late Month 2022-01-22  
1                Medium          0  Late Month 2023-01-21  
2                Medium          0  Early Month 2021-01-06  
3                 High          1  Early Month 2023-01-06  
4                 High          1  Late Month 2023-01-23  
  
df.to_csv("cleaned_cybercrime_dataset.csv", index=False)  
print("Dataset exported successfully")  
Dataset exported successfully.
```

Step 10 – Final Validation and Export

DATA EXPORT TO SQL SERVER

After cleaning and feature engineering, the final dataset was exported and used in SQL Server. SQL analysis was performed to generate structured insights such as executive summaries, yearly trends, attack-type risks, city-level losses, sector-wise losses, and high-risk attack detection.

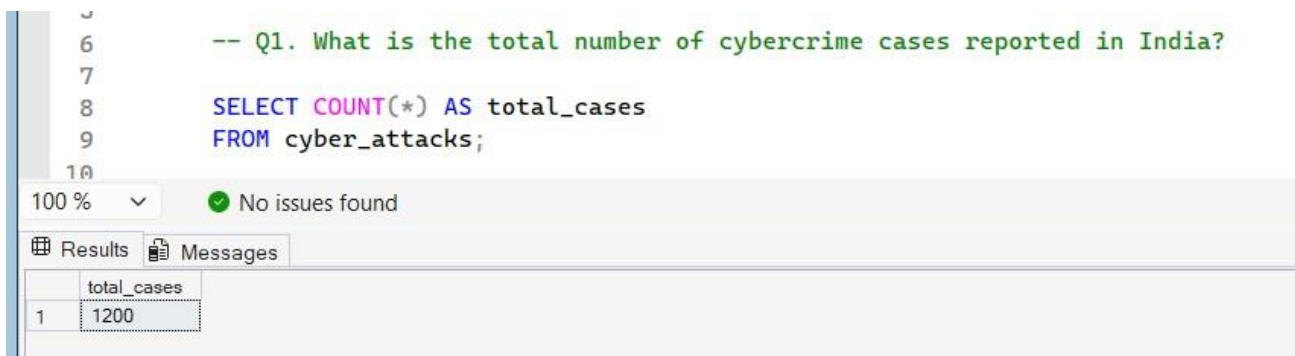
Database Table Name: cyber_attacks
Database Name: cybercrime_analytics_db

DATA ANALYSIS USING SQL SERVER

SQL Server was used as the analytical layer of the project. The queries were designed to answer real cybersecurity intelligence questions and support Power BI dashboard visuals.

Q1 – Total Number of Cybercrime Cases

The result shows 1200 total cybercrime cases in the dataset.



The screenshot shows a SQL query in a text editor and its results in a table. The query is: `-- Q1. What is the total number of cybercrime cases reported in India? SELECT COUNT(*) AS total_cases FROM cyber_attacks;` The results table has one column, `total_cases`, and one row with the value 1200. The interface also shows a status bar with 'No issues found' and tabs for 'Results' and 'Messages'.

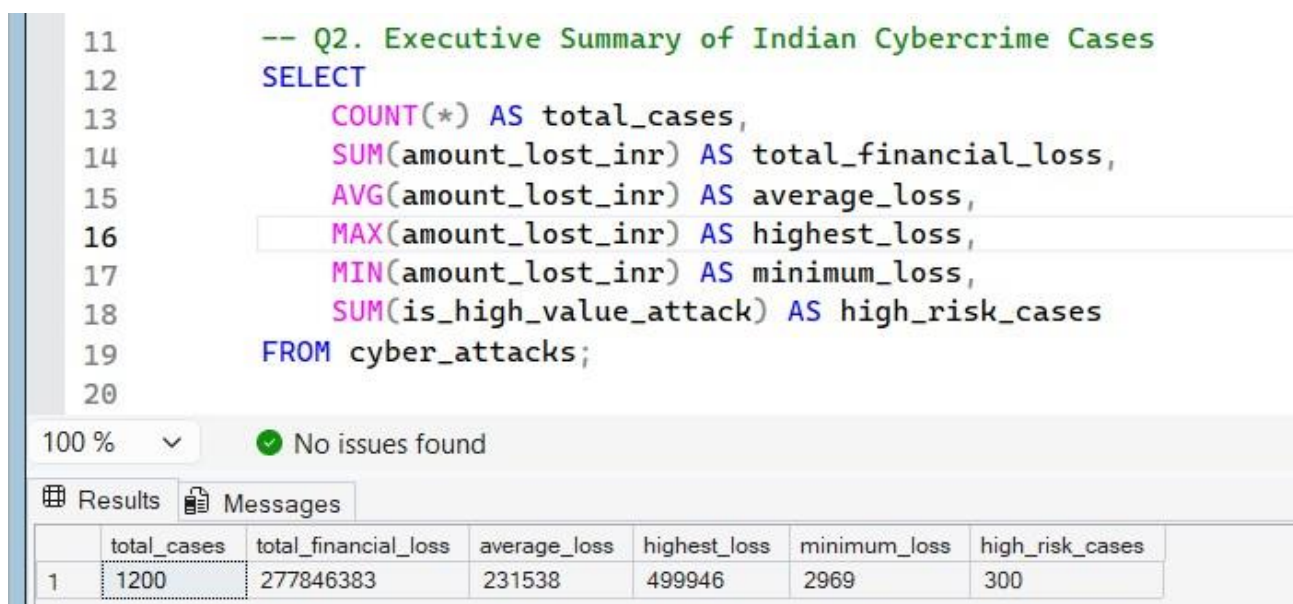
```
-- Q1. What is the total number of cybercrime cases reported in India?
SELECT COUNT(*) AS total_cases
FROM cyber_attacks;
```

	total_cases
1	1200

Q1 – Total Number of Cybercrime Cases

Q2 – Executive Summary of Indian Cybercrime Cases

This query calculates total cases, total financial loss, average loss, highest loss, minimum loss, and high-risk cases.



The screenshot shows a SQL query in a text editor and its results in a table. The query is: `-- Q2. Executive Summary of Indian Cybercrime Cases SELECT COUNT(*) AS total_cases, SUM(amount_lost_inr) AS total_financial_loss, AVG(amount_lost_inr) AS average_loss, MAX(amount_lost_inr) AS highest_loss, MIN(amount_lost_inr) AS minimum_loss, SUM(is_high_value_attack) AS high_risk_cases FROM cyber_attacks;` The results table has six columns: `total_cases`, `total_financial_loss`, `average_loss`, `highest_loss`, `minimum_loss`, and `high_risk_cases`. The first row contains the values 1200, 277846383, 231538, 499946, 2969, and 300. The interface also shows a status bar with 'No issues found' and tabs for 'Results' and 'Messages'.

```
-- Q2. Executive Summary of Indian Cybercrime Cases
SELECT
    COUNT(*) AS total_cases,
    SUM(amount_lost_inr) AS total_financial_loss,
    AVG(amount_lost_inr) AS average_loss,
    MAX(amount_lost_inr) AS highest_loss,
    MIN(amount_lost_inr) AS minimum_loss,
    SUM(is_high_value_attack) AS high_risk_cases
FROM cyber_attacks;
```

	total_cases	total_financial_loss	average_loss	highest_loss	minimum_loss	high_risk_cases
1	1200	277846383	231538	499946	2969	300

Q2 – Executive Summary of Indian Cybercrime Cases

Q3 – Yearly Cybercrime Trend and Financial Growth

This query compares yearly case counts, total financial losses, and high-risk cases from 2019 to 2024.

```
21      -- Q3. Yearly Cybercrime Trend & Financial Growth Analysis
22      SELECT
23          year,
24          COUNT(*) AS total_cases,
25          SUM(amount_lost_inr) AS total_financial_loss,
26          SUM(is_high_value_attack) AS high_risk_cases
27      FROM cyber_attacks
28      GROUP BY year
29      ORDER BY year;
30
```

100 % No issues found

Results Messages

	year	total_cases	total_financial_loss	high_risk_cases
1	2019	181	44472964	54
2	2020	204	44321050	41
3	2021	194	45315666	53
4	2022	214	50977080	53
5	2023	212	47654552	47
6	2024	195	45105071	52

Q3 – Yearly Cybercrime Trend and Financial Growth

Q4 – Cyber Attack Type Analysis with Risk Evaluation

Ransomware, malware, phishing, data breach, and online fraud are analyzed by total cases, financial loss, and high-risk cases.

```
31      -- Q4. Cyber Attack Type Analysis with Risk Evaluation
32      SELECT
33          incident_type,
34          COUNT(*) AS total_cases,
35          SUM(amount_lost_inr) AS total_financial_loss,
36          MAX(amount_lost_inr) AS highest_loss,
37          SUM(is_high_value_attack) AS high_risk_cases
38      FROM cyber_attacks
39      GROUP BY incident_type
40      ORDER BY total_financial_loss DESC;
```

100 % No issues found

Results Messages

	incident_type	total_cases	total_financial_loss	highest_loss	high_risk_cases
1	Ransomware	189	41098786	499172	39
2	Malware	158	36082417	497758	36
3	Phishing	156	34713968	489096	38
4	Data Breach	136	34102131	494948	38
5	Online Fraud	153	33822815	498660	40
6	Hacking	131	32620076	498415	36
7	Identity Theft	117	31035885	499090	41
8	Cyber Bullying	116	28154376	499946	32
9	Others	44	6215929	294461	0

Q4 – Cyber Attack Type Analysis with Risk Evaluation

Q5 – Indian City-Level Cybercrime Risk Analysis

Delhi, Mumbai, Chennai, Bangalore, and Kolkata are identified as major cybercrime financial loss cities.

```
42      -- Q5. Indian City-Level Cybercrime Risk Analysis
43      SELECT
44          city,
45          COUNT(*) AS total_cases,
46          SUM(amount_lost_inr) AS total_financial_loss,
47          SUM(is_high_value_attack) AS high_risk_cases
48      FROM cyber_attacks
49      GROUP BY city
50      ORDER BY total_financial_loss DESC;
51
```

100 % No issues found

Results Messages

	city	total_cases	total_financial_loss	high_risk_cases
1	Delhi	143	33992810	36
2	Mumbai	155	33712006	37
3	Chennai	142	33476750	34
4	Bangalore	135	29669166	30
5	Kolkata	105	27633644	35
6	Lucknow	128	27432655	28
7	Pune	102	23785641	25
8	Jaipur	101	23463446	24
9	Ahmedabad	93	23021139	30
10	Hyderabad	96	21659126	21

Q5 – Indian City-Level Cybercrime Risk Analysis

Q6 – Sector-Wise Cybercrime Threat Intelligence

Social Media, Corporate, Financial, Government, and Personal sectors are analyzed by financial loss and high-risk cases.

```
52      -- Q6. Sector-Wise Cybercrime Threat Intelligence Analysis
53      SELECT
54          category,
55          COUNT(*) AS total_cases,
56          SUM(amount_lost_inr) AS total_financial_loss,
57          SUM(is_high_value_attack) AS high_risk_cases
58      FROM cyber_attacks
59      GROUP BY category
60      ORDER BY total_financial_loss DESC;
61
```

100 % No issues found

Results Messages

	category	total_cases	total_financial_loss	high_risk_cases
1	Social Media	177	38903406	41
2	Corporate	176	38513556	37
3	Financial	152	35560186	41
4	Government	154	34910952	39
5	Personal	159	34621260	28
6	Health	133	33328474	40
7	Educational	127	32073146	40
8	E-Commerce	122	29935403	34

Q6 – Sector-Wise Cybercrime Threat Intelligence

Q7 – High-Risk Cyber Attack Detection

This query filters high-value attacks and groups them by incident type, city, and category.

```
62      -- Q7. High-Risk Cyber Attack Detection Analysis
63      SELECT
64          incident_type,
65          city,
66          category,
67          COUNT(*) AS high_risk_cases,
68          SUM(amount_lost_inr) AS total_high_risk_loss
69      FROM cyber_attacks
70      WHERE is_high_value_attack = 1
71      GROUP BY incident_type, city, category
72      ORDER BY total_high_risk_loss DESC;
```

100 % No issues found

	incident_type	city	category	high_risk_cases	total_high_risk_loss
1	Cyber Bullying	Bangalore	Government	3	1291775
2	Malware	Bangalore	Educational	3	1266484
3	Cyber Bullying	Delhi	Financial	3	1262421
4	Online Fraud	Hyderabad	Health	3	1230384
5	Hacking	Mumbai	Health	3	1218937
6	Identity Theft	Delhi	Financial	3	1185736
7	Identity Theft	Ahmedabad	Financial	2	978921
8	Cyber Bullying	Chennai	Personal	2	956914
9	Hacking	Hyderabad	Social Media	2	943783
10	Phishing	Chennai	Government	2	940070
11	Phishing	Pune	Financial	2	919295
12	Phishing	Ahmedabad	Health	2	916872
13	Ransomware	Bangalore	E-Commerce	2	907598
14	Ransomware	Kolkata	Corporate	2	905191
15	Identity Theft	Mumbai	E-Commerce	2	898555
16	Phishing	Mumbai	Government	2	892386
17	Data Breach	Pune	Government	2	888198
18	Malware	Lucknow	Corporate	2	885763
19	Data Breach	Kolkata	Government	2	884003
20	Phishing	Delhi	Educational	2	875378
21	Online Fraud	Kolkata	Financial	2	872290
22	Cyber Bullying	Delhi	Social Media	2	871954
23	Malware	Ahmedabad	Government	2	871040

Q7 – High-Risk Cyber Attack Detection

Q8 – Monthly Period-Based Cybercrime Pattern

Early Month and Mid Month show higher cybercrime activity compared with Late Month.

```
74      -- Q8. Monthly Period-Based Cybercrime Pattern Analysis
75      SELECT
76          day_range,
77          COUNT(*) AS total_cases,
78          SUM(amount_lost_inr) AS total_financial_loss
79      FROM cyber_attacks
80      GROUP BY day_range
81      ORDER BY total_financial_loss DESC;
```

100 % No issues found

	day_range	total_cases	total_financial_loss
1	Early Month	433	99850040
2	Mid Month	428	96355216
3	Late Month	339	81641127

Q8 – Monthly Period-Based Cybercrime Pattern

Q9 – Top 10 Highest Financial Loss Incidents

This table identifies individual cybercrime incidents with the highest recorded losses.

```
83 -- Q9. Top 10 Highest Financial Loss Cybercrime Incidents
84 SELECT TOP 10
85     year,
86     city,
87     incident_type,
88     category,
89     amount_lost_inr
90 FROM cyber_attacks
91 ORDER BY amount_lost_inr DESC;
92
```

100 % No issues found

Results Messages

	year	city	incident_type	category	amount_lost_inr
1	2023	Ahmedabad	Cyber Bullying	Educational	499946
2	2022	Bangalore	Cyber Bullying	Government	499500
3	2020	Bangalore	Ransomware	E-Commerce	499172
4	2022	Mumbai	Identity Theft	E-Commerce	499090
5	2024	Chennai	Online Fraud	Social Media	498660
6	2024	Chennai	Hacking	Social Media	498415
7	2019	Pune	Malware	Educational	497758
8	2019	Ahmedabad	Hacking	Government	497037
9	2022	Delhi	Cyber Bullying	Corporate	496370
10	2020	Bangalore	Identity Theft	Personal	495509

Q9 – Top 10 Highest Financial Loss Incidents

Q11 – Top Cybercrime Affected Indian Cities Ranking

DENSE_RANK was used to rank cities by total financial loss.

```
92
93 -- Q11. Top Cybercrime Affected Indian Cities Ranking
94 SELECT
95     city,
96     COUNT(*) AS total_cases,
97     SUM(amount_lost_inr) AS total_financial_loss,
98     DENSE_RANK() OVER (
99         ORDER BY SUM(amount_lost_inr) DESC
100     ) AS city_risk_rank
101 FROM cyber_attacks
102 GROUP BY city;
```

100 %

Results Messages

	city	total_cases	total_financial_loss	city_risk_rank
1	Delhi	143	33992810	1
2	Mumbai	155	33712006	2
3	Chennai	142	33476750	3
4	Bangalore	135	29669166	4
5	Kolkata	105	27633644	5
6	Lucknow	128	27432655	6
7	Pune	102	23785641	7
8	Jaipur	101	23463446	8
9	Ahmedabad	93	23021139	9
10	Hyderabad	96	21659126	10

Q11 – Top Cybercrime Affected Indian Cities Ranking

Q12 – Year-over-Year Financial Loss Growth Analysis

The LAG function compares current-year loss with previous-year loss.

```
104      -- Q12. Year-over-Year Financial Loss Growth Analysis
105      SELECT
106          year,
107          SUM(amount_lost_inr) AS yearly_loss,
108          LAG(SUM(amount_lost_inr))
109          OVER (ORDER BY year) AS previous_year_loss
110      FROM cyber_attacks
111      GROUP BY year
112      ORDER BY year;
```

	year	yearly_loss	previous_year_loss
1	2019	44472964	NULL
2	2020	44321050	44472964
3	2021	45315666	44321050
4	2022	50977080	45315666
5	2023	47654552	50977080
6	2024	45105071	47654552

Q12 – Year-over-Year Financial Loss Growth Analysis

Q13 – Comprehensive Threat Intelligence Summary Report

This final SQL query combines year, city, incident type, category, loss category, total cases, financial loss, and high-risk cases.

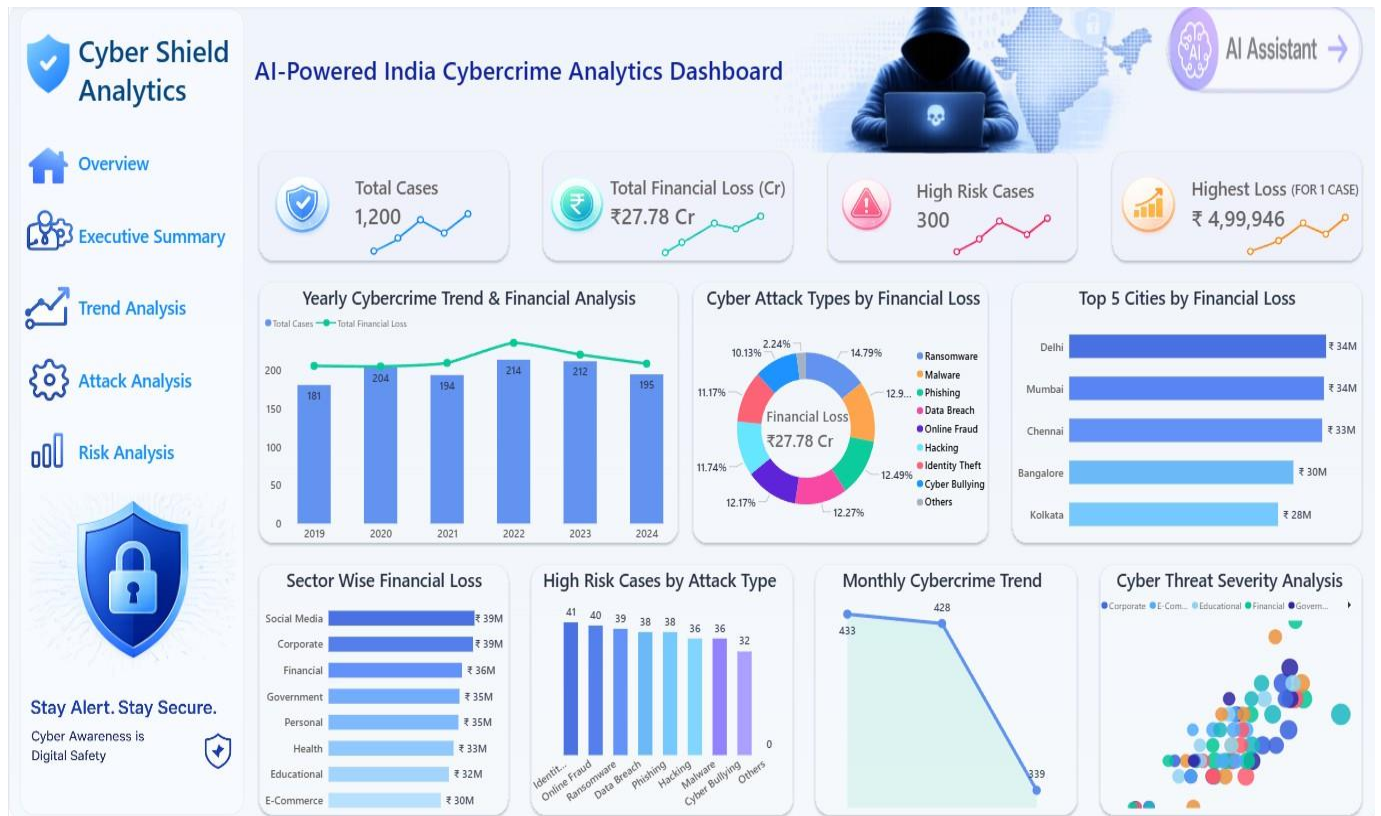
```
114      -- Q13. Comprehensive Threat Intelligence Summary Report
115      SELECT
116          year,
117          city,
118          incident_type,
119          category,
120          amount_lost_category,
121          COUNT(*) AS total_cases,
122          SUM(amount_lost_inr) AS total_financial_loss,
123          SUM(is_high_value_attack) AS high_risk_cases
124      FROM cyber_attacks
125      GROUP BY
126          year,
127          city,
128          incident_type,
129          category,
130          amount_lost_category
131      ORDER BY total_financial_loss DESC;
```

	year	city	incident_type	category	amount_lost_category	total_cases	total_financial_loss	high_risk_cases
1	2022	Delhi	Ransomware	Personal	Medium	6	1342574	0
2	2022	Delhi	Malware	Personal	Medium	5	1247154	0
3	2023	Lucknow	Ransomware	Government	Medium	5	1224782	0
4	2024	Chennai	Others	Social Media	Medium	5	1167631	0
5	2024	Chennai	Ransomware	Social Media	Medium	5	1109425	0
6	2024	Kolkata	Hacking	Financial	High	3	1073001	2
7	2021	Bangalore	Malware	Corporate	Medium	5	1059738	0
8	2020	Hyderabad	Hacking	Social Media	High	2	943783	2
9	2021	Bangalore	Others	Corporate	Medium	4	940334	0
10	2020	Bangalore	Ransomware	E-Commerce	High	2	907598	2
11	2022	Mumbai	Identity Theft	E-Commerce	High	2	898555	2
12	2022	Delhi	Phishing	Educational	High	2	875378	2
13	2019	Ahmedabad	Malware	Government	High	2	871040	2
14	2022	Delhi	Phishing	Personal	Medium	4	870174	0
15	2022	Mumbai	Others	E-Commerce	High	2	866688	2

Q13 – Comprehensive Threat Intelligence Summary Report

POWER BI DASHBOARD DEVELOPMENT

An interactive Power BI dashboard was designed using a futuristic cyber-intelligence theme. The dashboard includes a custom sidebar, KPI cards, sparklines, cyber-themed visuals, attack analysis, city and sector analysis, and a Streamlit AI assistant button.



Final Power BI dashboard for AI-Powered India Cybercrime Analytics

KPI Cards Used

KPI	Value	Meaning
Total Cases	1,200	Total cybercrime incidents in dataset
Total Financial Loss	■ 27.78 Cr	Total loss caused by cybercrime cases
High Risk Cases	300	High-value/high-risk attack records
Highest Loss	■ 4,99,946	Highest single financial loss recorded

Dashboard Visuals

Visual	Purpose
Yearly Cybercrime Trend & Financial Analysis	Compares yearly case count and financial loss
Cyber Attack Types by Financial Loss	Shows contribution of each attack type
Top 5 Cities by Financial Loss	Identifies highest-risk cities
Sector-Wise Financial Loss	Shows affected sectors by financial loss
High Risk Cases by Attack Type	Compares severe attacks by type
Monthly Cybercrime Trend	Shows early/mid/late month patterns
Cyber Threat Severity Analysis	Scatter plot showing risk distribution

AI ASSISTANT USING STREAMLIT

A Streamlit-based AI assistant called CyberInsight AI Assistant was developed as an additional intelligence layer. It is connected to the Power BI dashboard through a button, allowing users to open the assistant and ask questions about cybercrime insights.

CyberInsight AI Assistant
AI-powered assistant for Indian Cybercrime & Threat Intelligence Analytics

Metric	Value
Total Cases	1,200
Total Loss	₹27.78 Cr
High Risk Cases	300
Highest Loss	₹5.00 L

Quick Insights

- Top Risk City: Delhi
- Top Attack Type: Ransomware
- Highest Loss Year: 2022

Ask CyberInsight AI
Ask a question about your cybercrime dataset:
Example: Which city has the highest financial loss?
Generate AI Answer

CyberInsight AI Assistant built using Streamlit

AI Assistant Features

- Displays main KPIs such as total cases, total loss, high-risk cases, and highest loss.
- Shows quick insights such as top risk city, top attack type, and highest loss year.
- Provides an input box for asking cybercrime-related questions.
- Can be extended using Gemini, Groq, OpenRouter, or local Pandas-based responses.

PROJECT ARCHITECTURE

Layer	Description
Raw Dataset	Cybercrime incident records loaded from CSV
Python Cleaning	Data cleaning, datatype fixing, duplicate handling, and feature engineering
SQL Server	Advanced query analysis and threat intelligence summaries
Power BI Dashboard	Interactive analytics and visualization layer
Streamlit AI Assistant	AI-powered question answering and insight layer
Final Output	Cybercrime intelligence platform with dashboard and assistant

RESULTS AND DISCUSSION

- The dataset contains 1200 cybercrime records.
- Total financial loss is ■27.78 Cr.
- Ransomware is the highest financial loss attack type.
- Delhi, Mumbai, and Chennai are among the highest-risk cities.
- Social Media, Corporate, and Financial sectors show high cybercrime financial impact.
- The Power BI dashboard provides a complete visual summary of cybercrime intelligence.
- The AI assistant improves the project by adding an interactive query layer.

CONCLUSION

The AI-Powered Indian Cybercrime & Threat Intelligence Analytics Platform successfully demonstrates how raw cybercrime data can be transformed into meaningful insights using Python, SQL Server, Power BI, and AI integration. The project supports cybersecurity awareness, financial loss monitoring, city-level threat intelligence, sector-wise risk analysis, and AI-powered decision support.

FUTURE SCOPE

- Add real-time cybercrime data feeds.
- Use machine learning models for attack prediction.
- Add anomaly detection for unusual high-loss cases.
- Integrate APIs such as threat intelligence feeds.
- Improve AI assistant with live model APIs and natural language SQL.
- Deploy the Streamlit assistant online and connect from Power BI web link.

REFERENCES

- Python and Pandas documentation
- Microsoft SQL Server documentation
- Microsoft Power BI documentation
- Streamlit documentation
- Cybersecurity and threat intelligence references
- AI assistant integration documentation

END REPORT